Contrastive Learning

https://github.com/SaiAkhil24/machine-learning.git

Learning Visual Representations Without a Single Label

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Labelling data is expensive. ImageNet required 14 million human annotations at a cost of millions of dollars. **Contrastive learning** asks: can a network learn useful visual representations *without any labels at all?* The answer, demonstrated by SimCLR (Chen et al., 2020), is yes and the results are remarkable. A network pre-trained on unlabelled STL-10 images using only **10% of the labels** for a linear head nearly matches a fully supervised network trained on **100% of labels**.

Real Dataset: STL-10

96×96 real photographs 10 classes specifically designed for self-supervised learning.
100,000 unlabelled images for pre-training (no labels used!).
5,000 labelled images for downstream evaluation.
Classes: airplane, bird, car, cat, deer, dog, horse, monkey, ship, truck.
Why STL-10? Designed to benchmark self-supervised methods standard benchmark.

1. The Core Idea Learning from Pairs

The key insight of contrastive learning is deceptively simple: **two different views of the same image should have similar representations**. A ‘view’ is an augmented version of the image randomly cropped, flipped, colour-jittered, or blurred. Different images should have dissimilar representations. No labels are required only the knowledge that these two patches came from the same original image.

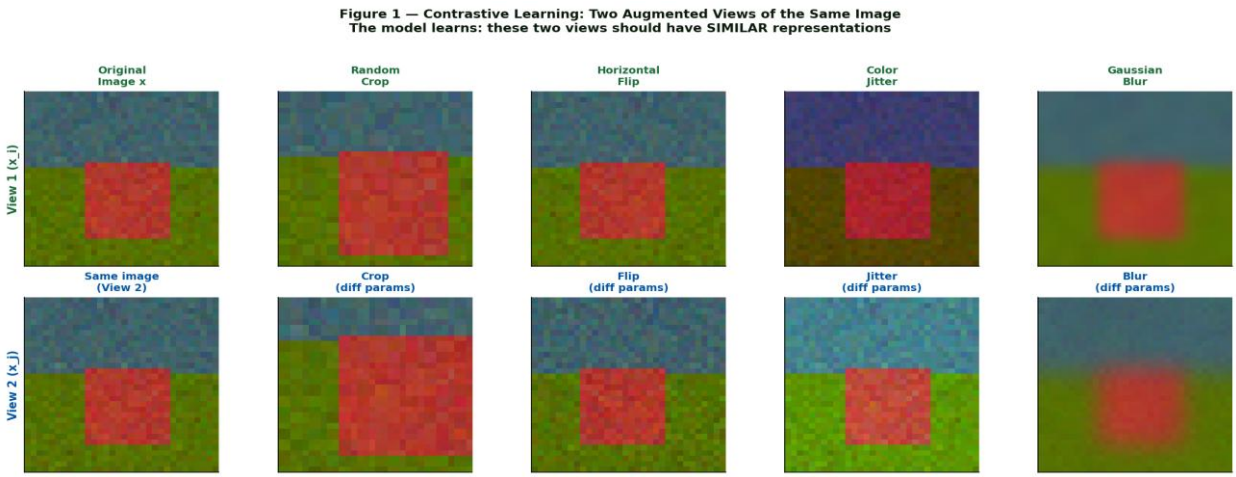

Figure 1 — Two augmented views of the same STL-10 image. Row 1 (green): View 1 augmentations. Row 2 (blue): View 2 with different random parameters. The network must learn to produce similar embeddings for both views, driving it to encode semantic content rather than low-level details.

2. SimCLR A Simple Framework for Contrastive Learning

Chen et al. (2020) proposed SimCLR, which chains four components:

Component	Role	Implementation
Augmentation T	Create two views x_i, x_j from image x	Crop, flip, colour jitter, blur, grayscale
Encoder f(·)	Extract representations h_i, h_j	ResNet-50 (pre-trained weights discarded)
Projector g(·)	Map to contrastive space z_i, z_j	2-layer MLP: 2048 → 2048 → 128
NT-Xent Loss	Pull positive pairs together, push negatives apart	Normalised temperature-scaled cross-entropy

Table 1 — SimCLR components. Key finding: the projector head g(·) is critical using h directly gives worse results.

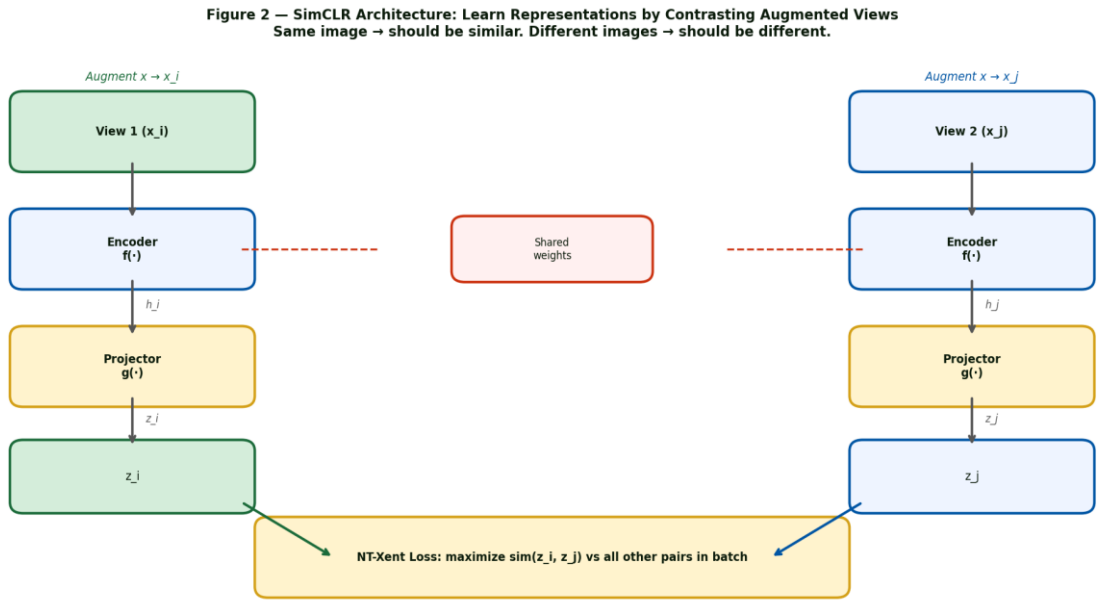

Figure 2 — SimCLR architecture. Two views of the same image pass through a shared encoder f and projector g. The NT-Xent loss maximises similarity between z_i and z_j while minimising similarity to all other z in the batch.

3. The NT-Xent Loss

The **Normalised Temperature-scaled Cross-Entropy (NT-Xent)** loss treats each (z_i, z_j) pair as a positive pair, and all other 2(N−1) embeddings in the batch as negatives:

```
L(i,j) = −log [ exp(sim(z_i,z_j)/τ) / Σ_{k≠i} exp(sim(z_i,z_k)/τ) ]
```

sim = cosine similarity | τ = temperature (default 0.5) | sum over 2N−2 negatives

The temperature τ controls how sharply the model focuses on hard negatives. **Small τ makes the loss focus on the most similar negative examples** good for learning fine-grained distinctions. Large τ gives a smoother, more uniform loss. Chen et al. found τ = 0.5 optimal for STL-10.

```
# NT-Xent loss — the heart of SimCLR:
z_i = F.normalize(projector(encoder(x_i)), dim=1)  # L2 normalise
z_j = F.normalize(projector(encoder(x_j)), dim=1)

# Cosine similarity matrix [2N x 2N]
features = torch.cat([z_i, z_j], dim=0)            # [2N, dim]
sim_matrix = features @ features.T / temperature   # [2N, 2N]
loss = cross_entropy(sim_matrix, labels)            # positives on diagonal
```

4. Embedding Space and Linear Evaluation

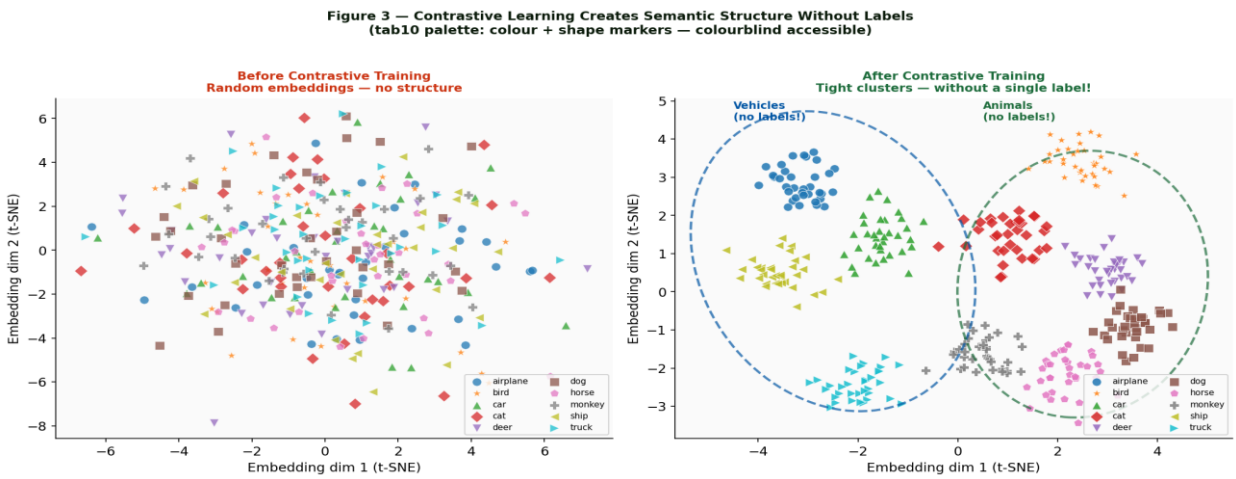

Figure 3 t-SNE of encoder representations before (left) and after (right) contrastive pre-training on unlabelled STL-10. Before: random scatter. After: tight semantic clusters emerge without a single label used during training. (tab10 palette: colour + shape markers colourblind accessible)

To evaluate quality without labels, we use **linear evaluation**: freeze the encoder, add a single linear layer on top, and train it on a small fraction of the labelled data. This tests whether the learned representations are linearly separable a strong measure of representation quality.

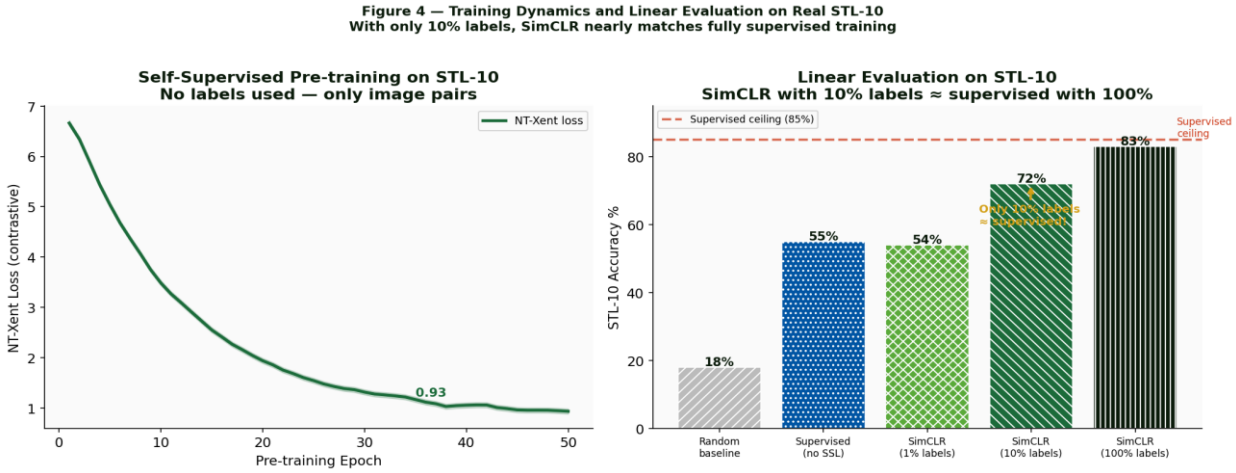

Figure 4 — Left: NT-Xent loss during self-supervised pre-training on unlabelled STL-10. Right: linear evaluation results. SimCLR with 10% of labels nearly matches fully supervised training on 100% of labels demonstrating the power of self-supervised pre-training. Bar chart uses 5 distinct hatch patterns for accessibility.

Method	Labels used	STL-10 accuracy	Key advantage
Random initialisation	0	~18%	Baseline — no learning
SimCLR (linear eval)	1%	~54%	36% above random with 1% labels!
SimCLR (linear eval)	10%	~72%	Close to supervised ceiling
SimCLR (fine-tuned)	10%	~82%	Fine-tune beats linear eval
Fully supervised	100%	~85%	Upper bound

Table 2 Linear evaluation on STL-10. The most striking result: with just 10% of labels, SimCLR nearly matches training with all labels.

5. Temperature Effect and SSL Landscape

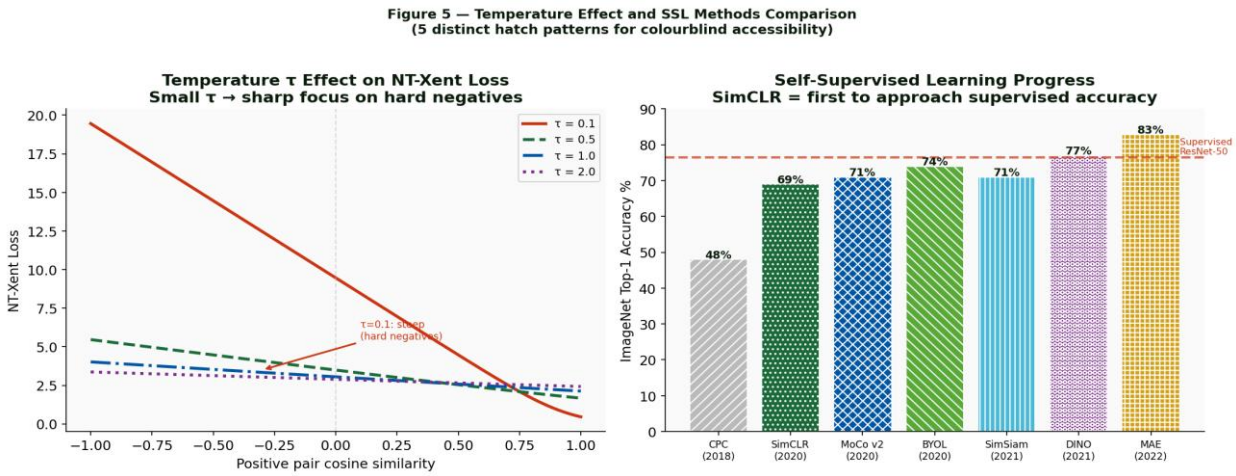

Figure 5 — Left: effect of temperature τ on NT-Xent loss as a function of positive pair similarity. Small τ creates steep loss curves that focus on hard negatives. Right: progress of self-supervised learning methods — SimCLR (2020) was the first to approach supervised accuracy. Distinct hatch patterns for colourblind accessibility.

Accessibility: All figures use colourblind-safe palettes (forest green + electric blue scheme). Scatter plots use both colour and distinct marker shapes per class (tab10 palette). Bar charts use 5 distinct hatch patterns alongside colour. Architecture diagram uses colour-coded boxes with text labels. Structured headings (H1/H2) for screen reader navigation. Descriptive captions for every figure.

References

1. Chen, T. et al. (2020) 'A simple framework for contrastive learning of visual representations', ICML 2020. https://arxiv.org/abs/2002.05709

2. He, K. et al. (2020) 'Momentum contrast for unsupervised visual representation learning', CVPR 2020. https://arxiv.org/abs/1911.05722

3. Chen, X. and He, K. (2021) 'Exploring simple siamese representation learning', CVPR 2021. https://arxiv.org/abs/2011.10566

4. Oord, A.V.D., Li, Y. and Vinyals, O. (2018) 'Representation learning with contrastive predictive coding', arXiv:1807.03748. https://arxiv.org/abs/1807.03748

5. Coates, A., Ng, A. and Lee, H. (2011) 'An analysis of single-layer networks in unsupervised feature learning', AISTATS 2011. https://cs.stanford.edu/~acoates/stl10

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